

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourgish Professional Newsroom

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotator-population mismatch is the highest-confidence concern in this analysis. Sentiment analysis annotation achieved Cohen's Kappa of only 0.45 [\[Q27\]](#) — below the 0.60–0.80 range typical for robust NLU annotation — with disagreements resolved by annotators agreeing among themselves [\[Q28\]](#). Annotators beyond SA are described as 'student assistants' [\[Q114\]](#); no evidence of professional journalist or editor involvement, despite the deployment explicitly requiring journalistic-register ground truth [\[A3\]](#). RTE labels are AI-assisted (ChatGPT-5.1 improvement [\[Q61\]](#), ChatGPT-5-MINI quality and label verification [\[Q63, Q64\]](#)), with manual correction only where the LLM altered semantics [\[Q65\]](#). JUDGEWEL NER uses LLMs as judges with minimal human verification [\[Q37\]](#). The benchmark itself warns 'models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices' [\[Q120\]](#). Given OC is a HIGH-priority dimension and the deployment requires journalist-register ground truth, score is 2.

ID	Evidence
Q27	"We calculated Cohen's Kappa at 0.45." (p.3)
Q28	"For the final set, the annotators agree on a label in cases of label disagreement." (p.3)
Q37	"The resulting sentences are then evaluated using LLMs acting as judges, with minimal human verification to calibrate quality thresholds." (p.4)
Q61	"Specifically, we first prompted CHATGPT-5.1 to assess and improve the translated sentence pairs unless they were already of very high quality, while explicitly keeping the original meaning to avoid label conflicts (see Appendix 7.4)." (p.5)
Q63	"To achieve (a), we prompted CHATGPT-5-MINI to judge the texts in the improved data and label their quality as either low, medium, or high, keeping only data rated at least medium, removing nearly 25% of the entire dataset (see Appendix 7.5)." (p.5)
Q64	"For (b), we prompted CHATGPT-5-MINI to verify whether the dataset labels remained correct after the first translation and improvement, outputting true or false for each sentence pair (see Appendix 7.6). Nearly 10% of the labels were false." (p.5)
Q65	"We found that the quality improvement step often corrected intentional logical contradictions or factual inaccuracies rather than keeping the original semantics. We therefore adjusted the sentences manually such that they corresponded to the ground truth again, while keeping false positives intact." (p.5)
Q114	"We would like to thank the student assistants for their annotation work." (p.9)
Q120	"As a result, models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices." (p.10)

Strengths

- The Lothritz et al. (2022) RTL NER subcorpus is fully human-annotated and high-precision [\[Q40, Q42\]](#)
- RTE labels underwent two-stage AI verification with manual correction where semantics drifted [\[Q62, Q65\]](#), offering some quality control
- The benchmark explicitly cautions deployers against treating performance as cultural/demographic coverage [\[Q121\]](#), enabling informed risk assessment

ID	Evidence
Q40	"The NER dataset introduced by Lothritz et al. (2022), by contrast, is a fully human-annotated corpus derived from RTL online news comments." (p.4)
Q42	"Annotation was conducted manually, yielding a smaller but high-precision dataset." (p.4)

Q62	"In addition, we perform two verification steps to make sure that (a) the quality of the improved texts is high enough and (b) that the labels are correct." (p.5)
Q65	"We found that the quality improvement step often corrected intentional logical contradictions or factual inaccuracies rather than keeping the original semantics. We therefore adjusted the sentences manually such that they corresponded to the ground truth again, while keeping false positives intact." (p.5)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels do not reflect professional Luxembourgish journalist/editor perspectives. The deployment requires the flexible journalistic register [A3], but annotators are described as student assistants [Q114] with no documented journalism background; LA labels derive from formal LOD/Spellchecker sources [Q30, Q31] and RTE labels are AI-assisted [Q63, Q64].

OC-2: Likely substantial disagreement between benchmark annotators and the target population (working journalists/editors), particularly on borderline LA Ortho cases [A3] where the deployment explicitly anticipates formal-vs-journalistic disagreement.

OC-3: Annotator demographics: SA was annotated by two native Luxembourgish speakers [Q25] with Cohen's Kappa 0.45 [Q27]; intent translation by one native speaker with consultation [Q52]; other tasks by 'student assistants' [Q114]. No professional-journalist annotators documented; no Datasheet-level demographic breakdown provided.

OC-4: Re-annotation of LA, HA, and a sample of NER by professional Luxembourgish editors applying the deployment's flexible register would be necessary to establish convergent validity for the deployment.

OC-5: SA disagreements resolved by annotators agreeing among themselves [Q28] — a consensus aggregation that erases minority annotator perspectives without documented arbitration. Cohen's Kappa 0.45 [Q27] indicates substantial baseline disagreement subsequently flattened.

OC-6: Documented label issues likely to harm convergent and external validity: (a) student-annotator pool [Q114] vs. professional-journalist target users [A3]; (b) AI-assisted RTE labels [Q63, Q64]; (c) LLM-as-judge JUDGEWEL [Q37]; (d) low SA IAA [Q27]; (e) explicit benchmark caution against demographic-coverage claims [Q120, Q121].

ID	Evidence
Q25	"In total, we extract 4,583 sentences, which are then annotated by two native speakers of LTZ." (p.3)
Q27	"We calculated Cohen's Kappa at 0.45." (p.3)
Q28	"For the final set, the annotators agree on a label in cases of label disagreement." (p.3)
Q30	"The sentences are derived from the Luxembourgish Online Dictionary (LOD) and are manipulated using the tags available in the dataset." (p.3)
Q31	"The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data." (p.3)

Q37	"The resulting sentences are then evaluated using LLMs acting as judges, with minimal human verification to calibrate quality thresholds." (p.4)
Q52	"The translations were performed by an LTZ native speaker. In cases of uncertainty, additional native LTZ speakers were consulted." (p.4)
Q63	"To achieve (a), we prompted CHATGPT-5-MINI to judge the texts in the improved data and label their quality as either low, medium, or high, keeping only data rated at least medium, removing nearly 25% of the entire dataset (see Appendix 7.5)." (p.5)
Q64	"For (b), we prompted CHATGPT-5-MINI to verify whether the dataset labels remained correct after the first translation and improvement, outputting true or false for each sentence pair (see Appendix 7.6). Nearly 10% of the labels were false." (p.5)
Q114	"We would like to thank the student assistants for their annotation work." (p.9)
Q120	"As a result, models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices." (p.10)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)
WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies (https://data.public.lu/en/organizations/zenter-fir-dletzebuerger-sprooch/)

Information Gaps

- Number, demographics, and journalism background of student annotators
- Whether any IAA was computed for tasks other than SA
- Adjudication procedure when SA annotators initially disagreed

Requires Expert Verification

- Disagreement rate between professional Luxembourgish editors and benchmark labels on a held-out sample, especially for LA Ortho and HA

Remediation

Gap 1: Annotators are student assistants [Q114] with no professional journalism background; SA Cohen's Kappa 0.45 [Q27] is below typical thresholds; RTE and JUDGEWEL labels are AI-assisted [Q37, Q63, Q64]

Recommendation: Re-annotate held-out HA and LA test sets with at least three professional Luxembourgish editors applying explicit journalistic-register guidelines; report IAA and adjudication procedure

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